Magnetization of the Spanish Peaks Dike Swarm, Colorado, and Shiprock Dike, New Mexico

E. E. LARSON

Department of Geological Sciences, University of Colorado, Boulder, Colorado 80302

D. W. STRANGWAY[1]

Department of Geology and Geophysics, Massachusetts Institute of Technology Cambridge, Massachusetts 02139

A detailed study of the magnetic properties of some of the radiating dikes from the Spanish Peaks, Colorado, and from a pair of dikes from Shiprock, New Mexico, indicates that there is no correlation between the direction of stable remanent magnetization and dike orientation. Some dikes have a stable component of remanent magnetization; others do not. In general, magnetic stability is correlated with the presence of small grains observed microscopically. In most samples ilmenite lamellas are few or entirely lacking, indicating that high-temperature oxidation has been minimal. However, most samples have been affected to some degree by low-temperature oxidation resulting in the alteration of titanomagnetite to titanomaghemite. There is no relation between the presence of maghemite and the degree of magnetic stability. The dikes with higher magnetic stability were used to determine a pole position for the Eocene of North America. This position is 81°N, 149°W, and is close to other lower Tertiary pole positions.

INTRODUCTION

Spanish Peaks, located in south-central Colorado (Figure 1), are considered to be the remnants of a Late Eocene or Early Oligocene volcano [*Johnson*, 1961; *Knopf*, 1936; *U. S. Geological Survey, Folio 71*]. The high central portion consisting of two peaks, referred to as the East and West Spanish Peaks (Figure 2), is the neck of the old vent. About fifty dikes ranging from 1 to 30 meters in width radiate outward from the central complex, for distances up to 15 km. The rocks range from granitic and syenitic to dioritic, indicating that the dikes were injected over a sufficient span of time for changes in the composition of the magma to occur. A later set of dikes that trend generally east–west are composed of basaltic or lamprophyric material.

The radiating and east–west dikes were sampled, primarily to test the hypothesis suggested by *Strangway* [1961] that the direction of thermoremanent magnetization (TRM) is controlled by the dike orientation. *Fahrig et al.* [1965], *deBoer* [1967], and *Evans* [1968] have presented strong evidence that the shape of the dike does not affect the direction of magnetization.

A pair of dikes was sampled at the diatreme plug at Shiprock, New Mexico. These dikes are also considered to be Tertiary but their precise age is unknown. During the course of the study the samples were examined in a variety of ways. Saturation magnetization versus temperature ($J_s - T$) and the magnetic stability to af demagnetization were measured. Studies of the minerals were made by microscopic observations of polished sections, by X-ray diffraction, and by wet chemical analysis. These examinations have provided useful data about the relation of rock magnetism and petrography.

PALEOMAGNETISM

The remanent magnetism of samples from fifteen dikes in the Spanish Peaks area (Figure 2) and two from Shiprock with varying strike directions was determined. In general, 5 to 10 samples taken from 4 or 5 sites in each dike were measured. The results of the measure-

[1] Now at Department of Physics, University of Toronto, Canada.

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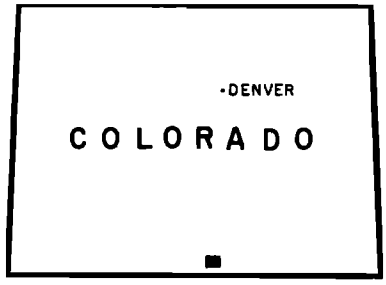


Fig. 1. Location of Spanish Peaks area, Colorado.

ment of natural remanent magnetization (NRM) are given in Tables 1, 2, and 3.

In the first part of Tables 1 and 3 are given the initial magnetic data from dikes 3, 5, 9, 14, 15, and 17 at Spanish Peaks and dike 20 at Shiprock; all eventually yielded well-grouped magnetic directions. The results of af demagnetization in peak fields up to 200 oe for these stable dikes are given in the first part of Table 2 and the second part of Table 3 and are shown in Figure 3. In general, after af demagnetization these dikes gave a precision parameter (k) of 18 or more and have a half-angle for the cone of 95% confidence (α_{95}) of less than 25°.

In the second part of Table 1 and the first part of Table 3 are given the NRM data from a second group of dikes (1, 2, 4, 8, 10, 12, 13, and 16 from Spanish Peaks and 19 from Shiprock), which after af demagnetization failed to give consistent, well-grouped directions of magnetization. Data for the unstable dikes after demagnetization are given in the second part of Table 2 and in Table 3. This group generally shows values of the precision parameter (k) less than about 18 and values of the cone of confidence (α_{95}) greater than 25° after demagnetization.

In summary, the dikes seem to fall into two categories, one magnetically stable and the other magnetically unstable. This finding is similar to findings reported by *Bruckshaw and Robertson* [1949] and by *Graham* [1953].

Pole positions for the stably magnetized dikes at Spanish Peaks (3, 5, 9, 14, 15, and 17) are included in Table 2 and have been plotted in Figure 4 with the ovals of confidence. The overall mean pole position, also shown in Figure 4, is statistically different from the present-day field and the rotational axis of the earth. (Shiprock data are not included in the mean, since these dikes are of unknown age.)

The dikes of Spanish Peaks include a variety of rock types, probably as a result of differentiation over a span of time. It seems reasonable, therefore, to suppose that the calculated mean pole position averages over enough time to approximate the paleomagnetic pole position

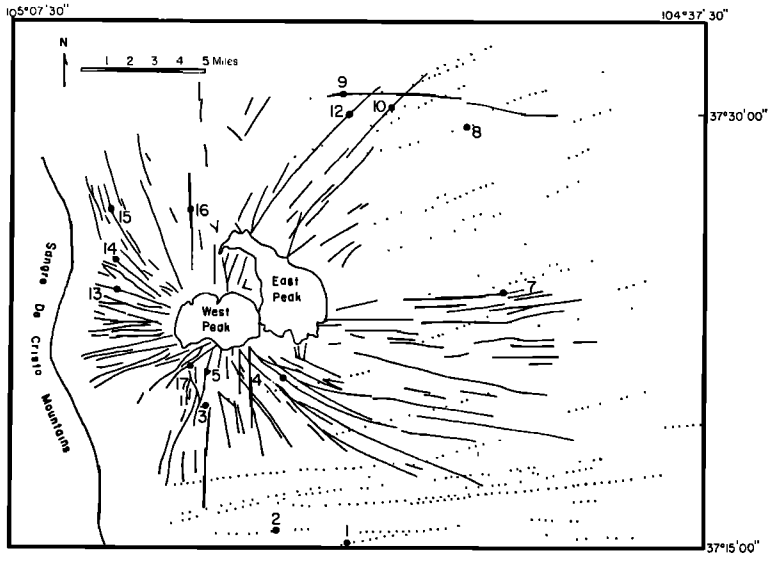


Fig. 2. Dikes in the Spanish Peaks area. Solid lines represent the radial dike swarm and dotted lines show a later east–west set. Numbers refer to sampling locations.

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TABLE 1. Paleomagnetic Data, Spanish Peaks before Demagnetization

Dike	No. of Sites	Total No. of Samples	D	I	K	α_{95}	Thickness, meters	Strike
			Dikes with High Magnetic Stability					
3	5	10	348	51	7.9	28.9	15	N 80°E
5	5	9	345	67	397.0	3.8	7	N 23°E
9	4	7	69	47	2.3	81.7	5	N 89°E
14	5	8	71	55	12.5	22.5	15	N 29°W
15	5	5	349	49	18.0	18.5	13	N 21°W
17	5	8	313	77	18.9	18.1	5	N 6°E
			Dikes with Low Magnetic Stability					
1	5	10	133	67	9.5	26.1	1.5	N 79°E
2	4	7	147	23	20.3	20.8	3.5	N 80°E
4	5	8	313	70	2.2	69.3	?	N 62°W
7	5	6	21	44	5.6	42.9	6	N 75°E
8	5	12	353	68	1.8	81.7	?	N 76°E
10	5	8	44	62	1.3	139.0	3	N 58°E
12	5	6	20	52	23.5	16.1	4	N 60°W
13	5	7	287	64	19.1	18.0	3	N 42°W
16	5	7	318	45	7.2	24.2	3	N 12°W

of the Late Eocene or Early Oligocene. Comparison of this position with other early Tertiary pole positions from North America is made in Figure 5 and in Table 4. The two positions *Irving* [1964] considers to be valid are the Siletz River volcanics [*Cox*, 1957] and the Green River formation [*Torreson et al.*, 1949]. Also included is the position for an early Tertiary intrusive from the Flagstaff intrusive in the Front Range near Boulder, Colorado [*McMahon and Strangway*, 1968]. The Spanish Peaks pole corresponds fairly closely with that from the Green River formation and the Front Range intrusive. The unusual pole position of the Siletz River volcanics is still unexplained.

TABLE 2. Paleomagnetic Data, Spanish Peaks after Demagnetization

	No. of Sites	Total No. of Samples	D	I	K	α_{95}	Pole Position				af Field
							Long.	Lat.	δM	δP	
			Dikes with High Magnetic Stability								
3	4	8	331	56	21.6	20.2	189	67	29	21	100
5	5	8	10	64	77.8	8.7	66	79	14	11	100
9	4	6	340	58	59.5	12.0	185	74	18	13	180
14	4	5	41	58	18.4	22.0	31	58	33	24	100
15	4	4	335	52	88.6	9.8	247	84	13	9	100
17	5	8	333	73	184.1	5.6	136	63	10	9	100
Mean	6		351	63	27.4	13.0	149	81	20	16	
			Dikes with Low Magnetic Stability								
1	5	8	132	74	12.5	22.6					100
2	4	5	159	56	8.8	32.8					200
4	5	6	283	38	2.2	68.6					150
8	5	13	295	54	2.2	67.9					200
10	5	7	87	31	3.9	44.5					200
12	5	5	342	66	1.9	79.2					100
13	3	5	31	68	18.2	42.3					150
16	4	4	76	58	1.6	124.1					200

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TABLE 3. Paleomagnetic Data, Shiprock

	No. of Sites	Total No. of Samples	D	I	K	α_{95}	Thickness, meters	Strike
Before Demagnetization								
19-Shiprock Dike	5	9	320	48	3.5	4.8	3	N 14°W
20-Shiprock Dike	5	9	333	68	43.3	11.8	1	N 64°E
After af Demagnetization								
							Pole Position	
							Long. Lat. δm δP	af Field
19	3	5	300	40	2.0	136.0	193 37 168 98	200
20	3	8	340	69	27.4	17.9	144 68 30 26	150

Correlation between Direction of Magnetization and Strike of Dike

Considering the directions of magnetization of the stably magnetized dikes and the strikes of those dikes (Figure 2, Tables 2 and 3), we conclude that there is no correlation between the orientation of the dike body and its direction of magnetization.

Rock Magnetism and Petrography

To assess the relations of the Fe-Ti phases present, a polished section was made from one sample of known remanence from each dike. These polished sections were observed under oil at magnifications up to 1000 and the grain sizes of opaques (both initial and any resulting from high-temperature oxidation changes) estimated. The phases present were also identified.

To further aid interpretation of the magnetic phases, curves of saturation magnetization versus temperature (J_s — T curves) were obtained for the same samples from which the polished sections were made.

Finally, these same samples were pulverized and the magnetic grains extracted. These grains were analyzed by X-ray diffraction, and their TiO_2-FeO-Fe_2O_3 proportions were obtained by chemical wet tests.

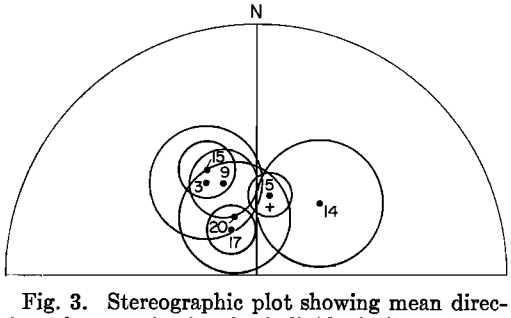


Fig. 3. Stereographic plot showing mean direction of magnetization for individual dikes after af demagnetization. Circles represent the 95% cones of confidence. Plus sign indicates the present direction of the magnetic field at this location.

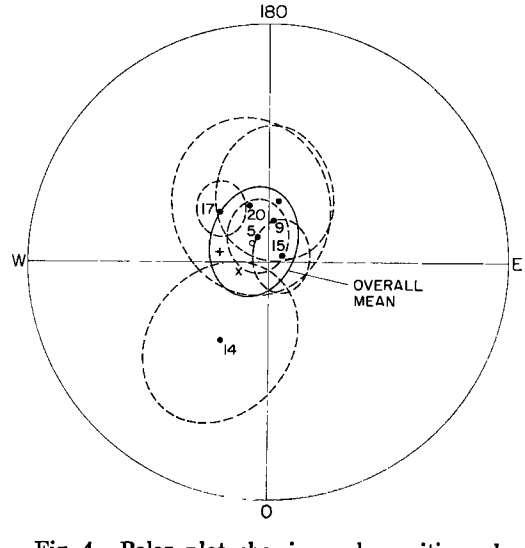


Fig. 4. Polar plot showing pole positions determined from individual dikes. Ovals represent the 95% confidence limits. Open circle is over-all mean pole position; plus sign is the present pole position; cross is the pole position determined from the present magnetic field in this area.

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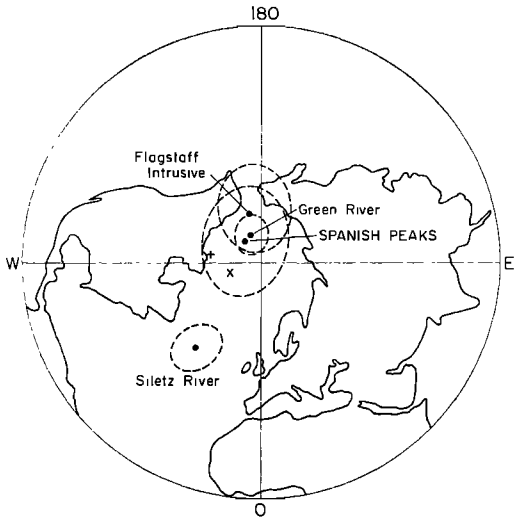


Fig. 5. Early Tertiary pole positions, North America, with 95% confidence limits. Plus sign is the present pole position; cross is the pole position determined from the present magnetic field in this area.

Table 5 summarizes the microscopic data. The k precision parameter is included to provide a rough correlation with the magnetic stability of the samples.

Stability of Magnetization versus Grain Size

Table 5 suggests that the samples that contain a relatively stable remanence are finer grained than the less stable samples. Commonly, the 'softer' samples, indicated by small k values, contained an abundance of quite large (up to 300 μ) grains. Because of the scarcity of high-temperature modifications due to the oxidation of grains by the development of ilmenite lamellas, most of the effective grain size is due to initial crystallization processes. (See *Larson et al.* [1969] for a discussion of the development of effective grain size.) As a check on the relation between grain size and stability, the magnetic data (k values) were compiled by one of us; the microscopic observations were conducted independently by the other. This led to a prediction of stability from microscopic data that could be compared with the magnetic data. In all cases except one, the two observations led to the same result. The exception was dike 14, which is only marginally stable magnetically.

Stability of Magnetization versus Maghemite Occurrence

Many of the samples studied show extensive low-temperature oxidation to maghemite. *Akimoto and Kushiro* [1960] concluded that the alteration of titanomagnetite to titanomaghemite greatly lowers the bulk coercivity of the sample. More recently, *Watkins* [1967] investigated some unstable basaltic rocks and concluded that their instability also was the result of their titanomaghemite content. The evidence from the present study indicates that grain size is the most important factor, whereas the titanomaghemite content is of subordinate importance. Samples 3-1, 5-4, 9-4, 17-2, and 20-1 have all been affected by extensive maghemitization, and yet they are among the more magnetically stable samples. On the other hand, stable samples 14-1 and 15-2 have essentially no microscopically discernible alteration to maghemite although chemical analysis reveals that these two samples are somewhat oxidized. Samples 1-2 and 2-3, which under the microscope also appear relatively unaltered, are magnetically unstable. There seems, therefore, to be no real correlation between magnetic stability and maghemitization. Because of the ubiquity of titanomaghemite alteration there arises the question of whether the magnetization was acquired during the initial cooling of the rocks or during subsequent alteration of the titanomagnetite.

Information about grain size and amount of ilmenite and hematite can be accurately determined optically, since these phases are readily identifiable under the reflecting microscope. The state of the titanomagnetite can, however, only be determined approximately. Where al-

TABLE 4. Lower Tertiary Pole Positions, North America

	Lat., Long. of Pole	δM	δP
Siletz River volcanics [11.098]*	37°, 49°W	7	10
Green River formation [11.100]*	78°, 158°W	6	7
Spanish Peaks, Colorado	81°, 149°W	16	20
Flagstaff intrusive, Front Range, Colo.	72°, 166°W	6	7

* Numbers in brackets refer to *Irving*, [1964].

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TABLE 5

Sample No.	Ilmenite: Grain Size, μ	Ilmenite: State	Magnetite: Initial Grain Size, μ	Magnetite: Resultant Grain Size, μ	High-Temp. Alteration (Exsolution)	Low-Temp. Alteration (Maghemite)	k
			With Stable Magnetization—Spanish Peaks				
3–1	10–40	Partly altered	<1–100 Ave. 20–30	Same	None	All grains wholly or partially altered	21.6
5–4	Rare	Altered to leucoxene	<1–100 Ave. 10 Many 1–5	Same	None	Small grains unaltered, large grains 80–90% altered	77.8
9–4	Rare		<1–100 Mainly 1–10	Same	None	Large and small grains 30–100% altered	59.5
14–1	Rare 5–40	Altered	<1–200 Ave. 10–15 Many <1	Same	None	Little altered	18.4
15–2	Common 5–30	Altered to leucoxene	<1–50 Ave. 10 μ Many < 1 μ	Same	None	Essentially unaltered	88.6
17–2	Common ?–30	Altered to leucoxene	<1–100 Ave. 15–20 Many <1	Same	None	All grains 50–100% altered	184.1
			No Stable Magnetization—Spanish Peaks				
1–2	50–80	Fresh	2–200 Ave. 30–50 Few small	2–50	Few broad ilmenite lamellas	Essentially unaltered, a little blue	12.5
2–3	Rare		1–100 Ave. 20–25 Few small	Same	None	Slightly altered	8.8
4–1	Rare		<1–300 Ave. 15–25 but many large grains	Same	None	95% altered	2.2
7–3	1–50	Fresh	3–150 Ave. 15–25	Same	Rare	Most grains corroded and pitted to some extent	No data
8–4	1–100	Fresh	1–150 Ave. 20–30 Many 1–10	Same	In 25% of large grains lamellas present	Most grains altered 50–100%	2.2
10–2	Rare		2–100 Ave. 15–30	Same	None	A few grains altered most slightly lightened	3.9
12.4	Rare		1–100 Ave. 20–30 Few small grains	Same	Lamellas in about 25% of grains	Highly altered 80–100%	1.9
13–1	Fairly rare 5–40	Altered	1–200 Ave. 20–35 Many large grains	Same	None	Slightly altered	18.2
16–1	Common 5–50	Most altered to leucoxene	<1–250 Ave. 20–30 Many <10 Few <1	Same	None	Highly altered large grains 80–100% small grains 50%	1.6
			Shiprock Dike				
19–1	Rare	Fresh	<1–50 Ave. 10–20 Few <1	Same	None	Fairly altered	2.0
20–1	Abundant	Fresh	<1–30 Abt. <1	Same	None	Completely altered	27.4

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teration to titanomaghemite is in its initial stages, it is commonly difficult to detect. In most cases, single-phase oxidation is associated with a progressive lightening in color of the grain, but, if this oxidation takes place uniformly and there are few unaffected grains for comparison, it is difficult to make the identification positively. Samples 1-2 and 2-3, for example, show little microscopic evidence of low-temperature oxidation but the inflated $J_s - T$ curves and the chemical analyses indicate that some oxidation has taken place.

When low-temperature oxidation is more advanced, it is readily apparent under the microscope, and the presence of maghemite can be verified by the $J_s - T$ curves and by X-ray diffraction. This advanced stage is represented by a distinct blue-gray color that begins to tint the lighter portions and the presence of irregular cracks (apparently due to volume change). The highest stages are associated with a spongy-looking material mottled by a ubiquitous black dust and spangled by red and yellow dew drops of hematite and rutile.

Curie Temperatures

Curie temperature analyses have been made on representative whole-rock crushed samples as illustrated in Figure 6. These analyses provide useful information for identifying the maghemitization. Grains that are low in titanium but highly maghemitized show a reduction in the Curie temperature and/or a significant loss in the saturation magnetization after heating (samples 4-1, 5-4, 12-4, 13-1, 16-1, and 17-2). On heating, the titanomaghemite is unstable and breaks down to weakly magnetic titanohematite resulting in a reduced saturation magnetization. The remaining titanomagnetite contains about the same proportion of titanium as the titanomaghemite [*Ozima and Larson*, 1967] and its Curie temperature is generally less than that of the initial titanomaghemite. Maghemite that is rich in titanium, on the other hand, decomposes at higher temperatures than the 600° or 700°C reached in these heating experiments. In these cases, the heating $J_s - T$ curve is reproducible on cooling and the Curie temperature is typically around 500°–600°C, which is higher than that expected for the titanomagnetite with a corresponding Ti-Fe ratio (sample (7-3).

In other cases, where titanomagnetite is the dominant phase, almost reproducible curves are obtained (1-2, 2-3, 10-2, 14-1, 15-2), and these curves cannot be distinguished from the curves of samples principally containing titanium-rich maghemite.

Chemical Analysis

Chemical analysis has been done on representative magnetic separates from the suite of dikes. Since this is a bulk analysis of separates, it should be recognized that the data represent averages and do not represent the full range of compositions and oxidation states that may in fact be present in the material. The data still provide useful information, and general conclusions about the minerals can be drawn. The data from this study superimposed on the Curie temperature-composition curves derived

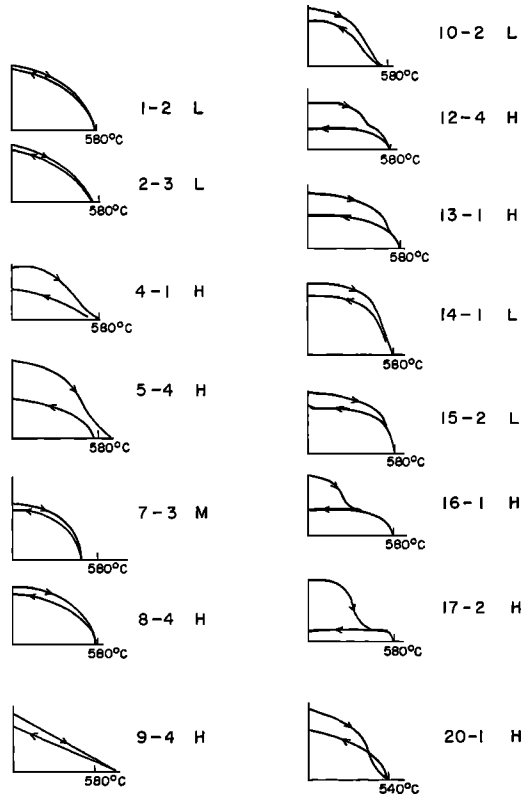


Fig. 6. Saturation magnetization versus temperature curves for selected whole-rock samples. L, M, and H refer to low, medium, and high states of oxidation, respectively, as observed microscopically.

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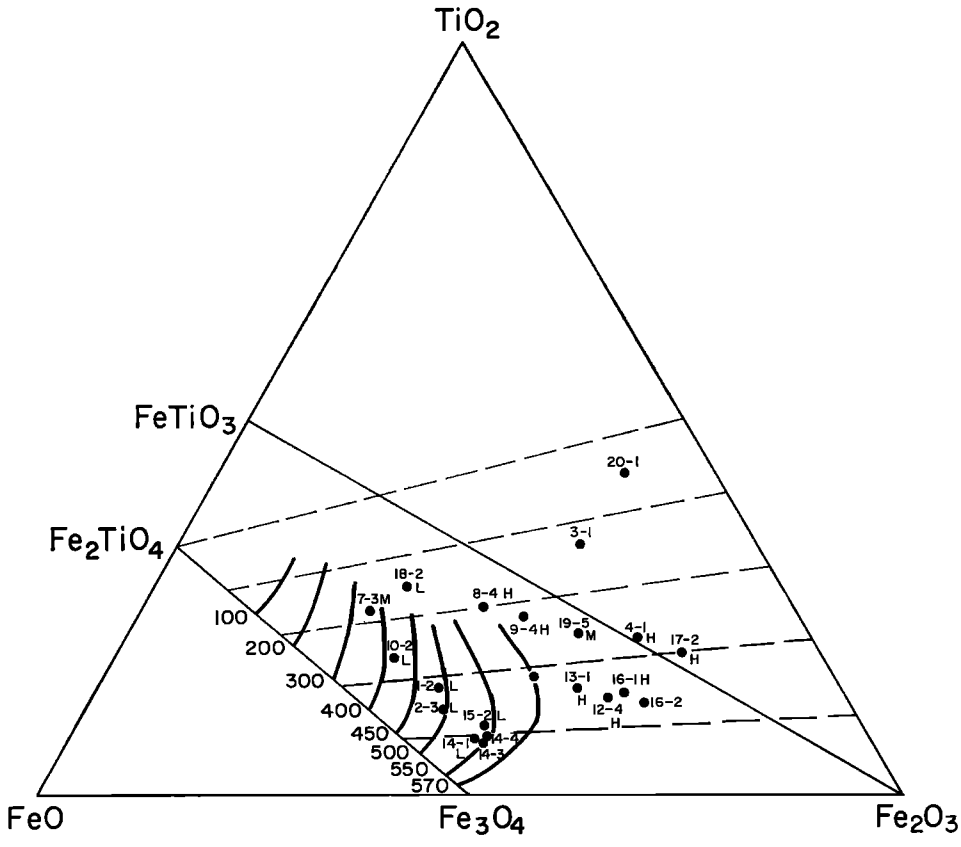


Fig. 7. Ternary diagram showing chemical composition of samples. Diagram shows lines of oxidation state (dashed) and Curie temperature curves. L, M, and H refer to low, medium, and high states of oxidation, respectively, as observed microscopically.

by *Akimoto et al.* [1957] are presented in Figure 7. Discrepancies between the measured Curie temperatures and the values predicted by the composition do exist but the deviations are not large. In general, it can be seen that the samples are highly oxidized, since the composition points are far from the Fe_3O_4-$FeTiO_3$ line. This corresponds with the presence of maghemite observed microscopically.

Many of the bulk separates have compositions that are more oxidized than the range for which the curves by Akimoto et al. are valid. This is partly due to the presence of large amounts of maghemite, although the bulk analysis may be somewhat misleading about the phases present. No precise estimates have been made of the per cent of maghemite present in the samples, but ranges of low (L), medium (M), and high (H) have been assigned (Figures 6 and 7), and it can be seen that the correlation between the chemical analysis and the microscopic observation is quite good. Thus the chemical analyses and the microscopic observations provide complimentary data that support considerable low-temperature oxidation of titanomagnetite to titanomaghemite.

X-Ray Diffraction Results

In addition to microscopic, thermomagnetic, and chemical studies of the maghemitization, an attempt was made to study the nature of the maghemite by measuring the lattice parameter. It is well known that the lattice parameter a of the titanomagnetite-ulvospinel solid solution series varies with titanium content and with the extent of maghemitization [*Nagata*, 1961]. With increased titanium content, a increases, whereas increasing oxidation tends to reduce

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the lattice parameter. A range of Ti compositions and oxidation states should tend to produce multiple X-ray peaks or diffraction-line broadening.

The Norelco diffractometer source was a Ni filtered radiation. The principal diffraction peak is between 35.0° and 36.0° (2θ), and this region was slowly scanned for each sample. Six representative diffractograms are reproduced in Figure 8. As can be seen, the main peak varied considerably in position and in width. The jaggedness and wideness of the peak undoubtedly reflect a range in composition A resulting from high- and low-temperature oxidation.

The lattice parameters calculated from the diffraction peak data are shown in Figure 9. The total variation in a is given by the bar width, whereas the principal lattice parameter range, corresponding to the highest sharpest peak on

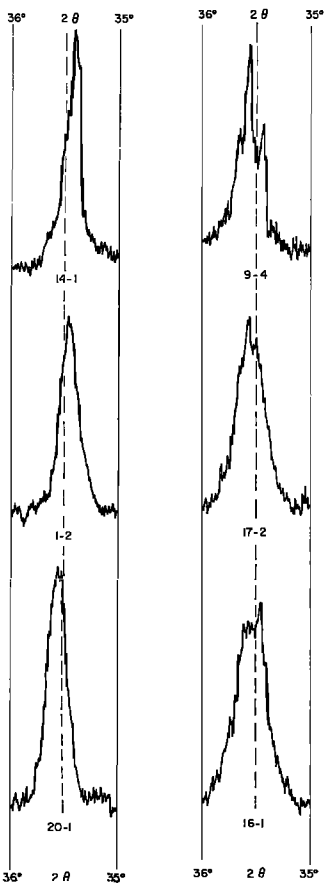


Fig. 8. X-ray diffractogram traces of representative samples.

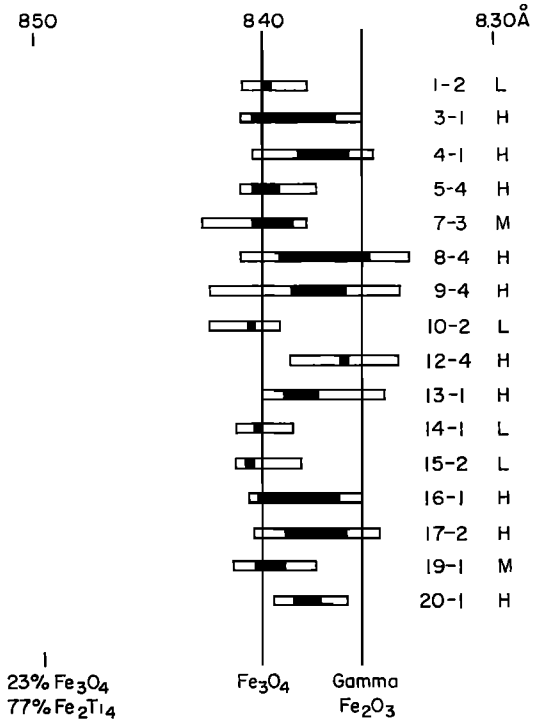


Fig. 9. Variation in lattice parameter a of magnetic fraction of samples. Range of a value shown by over-all length of bar and central flatter positions shown by shaded portions of bar.

the diffractogram, is indicated by the darker portion. If this figure is compared with Table 5, it is evident that there exists a correspondence between the X-ray results and the microscopic observations. Samples in which titanomaghemite is abundant show a wide range in lattice parameters and a tendency to have peaks between the peaks of magnetite and maghemite, whereas the samples that appeared little altered generally have rather narrow parameter limits and have peaks close to the pure magnetite value of 8.40 A. This seems to indicate that the magnetite is generally fairly titanium-free and that low-temperature oxidation is the main effect on the samples.

All samples showed a range of lattice parameters, however, indicating that all had undergone some high- or low-temperaure oxidation. The lattice parameter observations thus clearly complement the microscopic and chemical analyses and provide additional support for the extensive oxidation to maghemite.

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Conclusions

There is no relation between the direction of stable magnetization in a dike and the strike of the dike.

The calculated mean paleomagnetic pole position of the Spanish Peaks corresponds with that of the early Tertiary.

Magnetic stability has very little dependence on the presence or absence of titanomaghemite. Stability does, however, vary inversely with the size of the opaque grains and only samples abundant in grains with diameters less than 1 μ have a stable component of remanent magnetization.

X-ray diffractograms provide an additional means for estimating the range in composition (Ti content and variation in degree of oxidation).

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